A Semisupervised Infinite Latent Dirichlet Allocation Model for Target Discrimination in SAR Images With Complex Scenes

Lan Du, *Senior Member, IEEE*, Yan Wang, Weitong Xie, Zhaocheng Wang, and Jian Chen

***Abstract*— Synthetic aperture radar (SAR) target discrimination is usually performed in a supervised manner. However, supervised methods may suffer from lack of labeled training chips, whose acquirement is costly, time-consuming, and sometimes impossible. Moreover, traditional discrimination features only provide rough and partial description about chip and perform badly in SAR images with complex scenes. In order to solve these problems, we propose a novel semisupervised target discrimination method for SAR image by combining the feature learning with classifier learning into a uniform Bayesian framework based on a modified latent Dirichlet allocation (LDA) model, semisupervised infinite latent Dirichlet allocation (SSILDA). In our method, the semisupervised idea is used to deal with the difficulty of obtaining lots of labeled chips. A new variable is introduced into the LDA model for semisupervised learning, making it possible to obtain the semantic information of chips, while implementing the target discrimination in a semantic level. Moreover, the Dirichlet process (DP) is introduced into the LDA model to automatically determine the number of topics, and the parameters are inferred via Gibbs sampling. We have analyzed the performance of the proposed method comprehensively and specifically by using some measured data and carried out comparisons with the existing methods. The results validate the effectiveness of the proposed method for SAR target discrimination.**

***Index Terms*— Bayesian hierarchical model, Dirichlet process (DP), latent Dirichlet allocation (LDA) semisupervised learning, synthetic aperture radar (SAR), target discrimination.**

I. Introduction

SYNTHETIC aperture radar (SAR) has been widely used in both commercial and military areas, such as urban planning, natural resources monitoring, and military surveillance [1], due to its ability to form high-resolution images with relative invariance to weather and lighting conditions. With these advantages, SAR automatic target recognition (ATR) has also attracted much more attention during the past decades since it was proposed in 1993 [2]–[4]. Clemente *et al.* [3], [4] present ATR algorithms based on Krawtchouk moments and psedo-Zernike moments. A typical SAR ATR system often consists of three stages [5]–[8]: detection, discrimination, and classification or recognition. The detection stage searches the entire SAR image and provides the potential target regions, i.e., chips (small SAR images contain true targets or clutter false alarms), to the discrimination stage. The discrimination stage focuses on removing natural and artificial clutter chips and maintains the real targets chips. Finally, the classification stage classifies the remaining targets in the discrimination stage into different types. Since discrimination is the stage linking detection and classification/recognition, it directly influences the performance of the whole system [8]. As a result, SAR target discrimination has gained increasing attention during the last two decades [8].

Target discrimination is usually done by: 1) designing a set of features rich enough to discriminate targets from clutter and 2) learning a discriminator with these features from the training data. This strategy has been studied for many years and remarkable progress has been achieved.

In this strategy, features have a great influence on the discrimination performance. A great many of features have been presented during the last 20 years. For example, Lincoln Laboratory, Lexington, MA, USA, proposed the texture-based features in 1989 [9], including the fractal dimension, log standard deviation, and ranked fill ratio. In 1998, Verbout and Novak proposed the spatial boundary features [10], which captured the changes in spatial dispersion. Gao *et al.* [11] recently proposed the shape and contrast features for target discrimination in SAR images. These traditional features only provide rough and partial description about the target [12] and are more appropriate for discriminating targets from nature clutter in a simple scenario. However, due to ignoring the detailed shape, structure, and semantic information in chips, they cannot discriminate the target from clutter effectively in complex scenes. In fact, it can be observed that the targets of interest in SAR images, such as airplanes and vehicles, always have specific semantic concepts, which may be different from those of nature and artificial clutter. The limited understanding of the semantic coherence of chip in SAR image leads to a tremendous semantic gap for the target discrimination task. Consequently, exploitation of the semantic information contained in the chip is a promising way to improve the discrimination performance.

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Nowadays, a number of approaches have started to explore the spatial information by applying some low-level descriptors, such as scale-invariant feature transform (SIFT) [13], [14], histogram of oriented gradients (HOG) [15], and gray-level co-occurrence matrices [16], for remote sensing image (RSI) classification. Moreover, aiming at the multiplicative speckle noise in SAR images, Dellinger *et al.* [17] proposed the SAR-SIFT by defining the gradient of a pixel to be the log ratio of its neighbor regions. SAR-SIFT has been used in SAR target discrimination [12], [18]. However, these descriptors only catch limited local spatial geometric characteristics, which cannot be directly used to describe the structural contents of chips. Some articles have studied extracting mid-level features based on the statistic property of the descriptors to capture the structural information of the spatial region [19], [20]. Although to some extent these mid-level features can improve the classification accuracies, they cannot provide enough strong description and generalization ability for object classification in complex scenes. In recent years, probabilistic topic models, e.g., latent Dirichlet allocation (LDA) [21], which were originally proposed in text domain, have gained successful ground in natural image understanding [22], [23], as well as satellite image analysis [24], [25]. Under these models, each image is represented as a finite mixture over intermediate set of topics, which are expected to summarize the image semantic [26]. Some articles [24], [25], [27] have also showed that the classification or clustering results obtained by topic models can rely on semantic coherence. This research motivates us to consider building the high-level feature representation for SAR chips via LDA model to capture the semantic information contained in the chips.

The discriminator is another crucial factor for SAR target discrimination. The quadratic distance discriminator (QDD) [28] proposed by Lincoln Laboratory in 1993 is the first discriminator used for SAR target discrimination. It could achieve the optimal Bayes solution in theory. Kreithen *et al.* [29] applied QDD to the measured data and demonstrated that it can obtain good performance. Therefore, it has been widely used in many systems. However, the performance of the QDD decreases when mapping the n-dimensional feature space to the 1-D distance space according to the theoretical analysis [8]. Radisavljevic *et al.* [30] proposed the quadratic polynomial discriminator (QPD) in 1995. Though QPD is simple, it needs lots of samples to obtain the parameter vector and there is no theoretical method to choose the optimal kernel parameters. Moreover, only in the case that the data are assumed to be jointly Gaussian, QPD can achieve the minimum error rate [31]. Support vector machine (SVM) was proposed by Cortes and Vapnik in 1995 [32]. Due to its prominent performance, SVM rapidly becomes the main technology in machine learning and finds its applications in various fields. It has also been used in SAR target discrimination and gains pleasing results. However, most of the existing discriminators are supervised and their performances heavily rely on the amount of high-quality labeled samples. In practice, manual annotations procedure is time-consuming, difficult, and high cost, and may be faulty sometimes as we do not have enough information on the real scenes. Moreover, since the available labeled training samples are limited in SAR target discrimination, the underlying pattern of the image may not be properly captured. In order to solve these problems, we introduce the semisupervised learning into the SAR target discrimination.

The main idea of semisupervised learning is to jointly leverage the information of both labeled and unlabeled samples in the training process to improve the classification performance. Different semisupervised methods vary in the means of using labeled and unlabeled samples. Nowadays, the semisupervised methods presented in the literature can be grouped into the following main categories: 1) self-training; 2) co-training; 3) semisupervised SVM; 4) generative probabilistic models; and 5) graph-based semisupervised learning. Self-training is one of the earliest methods about using unlabeled data in the classification [33]–[35]. It starts by training classifier on the labeled samples only. Then, the most confident unlabeled samples, together with their predicted label, are used to augment the training set, and the classifier is retained with the augmented training set. This procedure is repeated until some criterion is reached. As a wrapper algorithm, self-training technique can be used with any supervised classifier. Co-training is proposed by Blum and Mitchell in 1998 [36]. It assumes that the input space can be divided into two sufficient and redundant views. Like the self-training algorithm, standard co-training works in an iterative manner. It iteratively trains the classifier and uses the prediction of the classifier on the unlabeled samples to augment the training set. Variants of co-training include multi-view learning [37] and multi-classifier learning [38]. Semisupervised SVMs are semisupervised learning techniques specially developed for SVM. Most of these algorithms use some type of regularization which encourages the fact that similar samples are associated to the same class. The effect of such regularization is to push the boundary between classes toward regions with low data density. Joachims [39] proposes a novel transductive SVM (TSVM) for RSI classification. TSVM gradually searches a reliable separating hyperplane with a transductive process that incorporates both labeled and unlabeled samples in the training phase. Generative probabilistic models are based on the estimation of the joint probability $p(x, y|\theta)$ assuming a particular model for the data, where θ is the parameter vector of the model that should be estimated from the observations. The estimation of the parameter vector θ can benefit from the joint exploitation of labeled and unlabeled samples. For unlabeled samples, the label y is regarded as the hidden variable. The final classification is then performed on the basis of the Bayes rule. The expectation–maximization (EM) algorithm is widely used to estimate θ. Tadjudin and Landgrebe [40] assume a Gaussian mixture model for the data and use the iterative EM algorithm to estimate model parameters from both labeled and unlabeled samples. Graph-based semisupervised learning methods define a graph where the nodes are samples, including labeled and unlabeled samples, and edges reflect their similarity. These methods usually assume label smoothness

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over the graph to include cluster/manifold regularization. Belkin and Niyogi [41] introduce an additional regularization term on the geometry of both labeled and unlabeled samples by using the graph Laplacian and propose the so-called Laplacian SVM (LapSVM). In [42], the LapSVM algorithm is applied to the classification of RSIs.

From the above analysis, we can see that in most of the existing semisupervised and supervised methods, the classifiers are designed independently from the features. In practice, it often occurs that the features are not suitable for the discriminator, in which the discrimination performance will be apparently degraded. Moreover, in these existing discrimination methods, the label information only contributes for the classifier learning. Actually, the label information can also be incorporated into the feature learning to improve the description ability of the features.

In order to address these issues mentioned above, in this article, we propose a novel semisupervised target discrimination method for SAR image based on the LDA model, where we combine the feature learning with classifier learning into a uniform Bayesian framework. First, the mid-level feature based on the bag of visual words (BoVW) model is extracted to represent the chip as sets of words. By doing this, we transform the chip into text-related terms, where a chip is viewed as a document and SAR-SIFT descriptors of patches extracted from the chip as the words. Then, a novel semisupervised infinite LDA (SSILDA) model, where we introduce a new class variable and the Dirichlet process (DP) into the original LDA, is trained with a few labeled samples and large amount of unlabeled samples. In this model, the SSILDA is used to analyze the word distribution between different patches of chips in order to obtain the topic distribution, which is the feature extracted in the semantic level. By introducing the LDA model, we project the original feature to the topic space and mine the underlying semantic information in chips. The new variable is regarded as the cluster label and can be used to indicate the label of chips. Moreover, the DP is used to automatically determine the number of topic. Finally, the test samples are put into the learned SSILDA to obtain their classes.

In summary, the main contributions of this article are threefold.

1) The semisupervised idea is introduced into SAR target discrimination to deal with the difficulty of obtaining large number of labeled samples.
2) We combine the feature learning and classifier learning into a uniform Bayesian framework by introducing a new variable into the LDA model. In this framework, the semantic information of chip is captured and the target discrimination is implemented in the semantic level.
3) We introduce the DP into the modified LDA model to automatically determine the number of topics.

The remainder of this article is organized as follows. In Section II, the original LDA is briefly introduced. Section III presents the proposed approach in detail. Experimental results and analysis are given in Section IV. Finally, the conclusion is drawn in Section V.

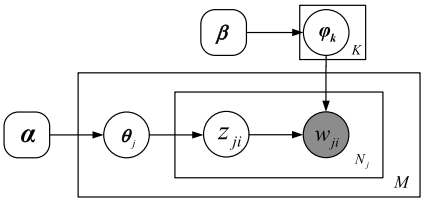


Fig. 1. Original LDA proposed in [21].

II. Original LDA and Its Mapping to Text

The LDA is first introduced in the text and has been extensively applied to analyze a variety of data, such as images [43], videos [44], and music [45]. In this article, we modify the LDA model to implement SAR target discrimination. In this section, we first review the original LDA model in the text analysis community and then map the chip into text for LDA modeling.

A. Original LDA

Fig. 1 shows the graphical model representation of LDA model in [21]. The boxes in Fig. 1 are "plates" representing replicates. The outer plate represents documents, while the inner plate represents the repeated choice of topics and words within a document. LDA is a generative probabilistic model of a set of documents, referred to as a corpus. The basic idea is that documents are represented as random mixtures over latent topics, where each topic is characterized by a distribution over words [21]. Suppose the corpus has M documents. Each document is modeled as a mixture of K topics, where K is assumed known. Each topic $k(k \in [1, \ldots, K])$ is modeled as a multinomial distribution $\boldsymbol{\beta}_k = [\beta_{k1}, \ldots, \beta_{kW}]$ over a word vocabulary of size W. $\boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_Q]$ is a Dirichlet prior over the word distribution on the topic. $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_K]$ is a Dirichlet prior over the topic distribution on the corpus. For each topic k, a word mixture $\boldsymbol{\varphi}_k$ is drawn from the Dirichlet distribution $\mathrm{Dir}(\boldsymbol{\varphi}_k|\boldsymbol{\beta})$. For each document j, a topic mixture $\theta_j = [\theta_{j1}, \ldots, \theta_{jK}]$ of the multinomial distribution over K topics is drawn from Dirichlet distribution $\mathrm{Dir}(\boldsymbol{\theta}_j|\boldsymbol{\alpha})$. For each word i in document j, a topic label $z_{ji} = k$ is drawn with probability θ_{jk}, and word w_{ji} is drawn from a discrete distribution given by $\varphi_{z_{ji}}$. $\boldsymbol{\theta}_j$ and z_{ji} are hidden variables. α and $\boldsymbol{\beta}$ are hyperparameters to be optimized.

The generative process of a document can be described by Algorithm 1.

The LDA model takes the occurrence counts of words, i.e., word frequency, in documents as input and then represents each document as a mixture over latent topics to summarize the document's semantic information [26]. In order to make it more easily to understand the LDA model, we show the topic representation process of LDA model in Fig. 2. Through the LDA model, we can obtain two kinds of information about documents: one is the distribution of the words over the topic, which is shared by all the documents, and the other one is the topic distribution of documents, which is varied with the document. The latter one can reflect the semantic information of documents. By the LDA model, the Q dimensional word

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Algorithm 1 Generative Process of the Documents in LDA

for each topic $k \in [1, \ldots, K]$ **do**
  Sample word mixture components $\boldsymbol{\varphi}_k \sim Dir(\boldsymbol{\beta})$
end for
for each document $j \in [1, \ldots, M]$ **do**
  Sample mixture proportion $\boldsymbol{\theta}_j \sim Dir(\boldsymbol{\alpha})$
  for each word $i \in [1, \ldots, N_j]$ in the document j **do**
    Sample topic index $z_{ji} \sim Mult(\theta_j)$
    Sample term for word $w_{ji} \sim Mult(\varphi_{z_{ji}})$
  end for
end for

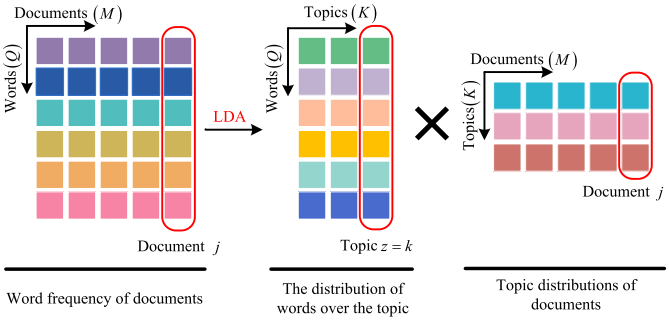


Fig. 2. Topic representation process of the LDA model.

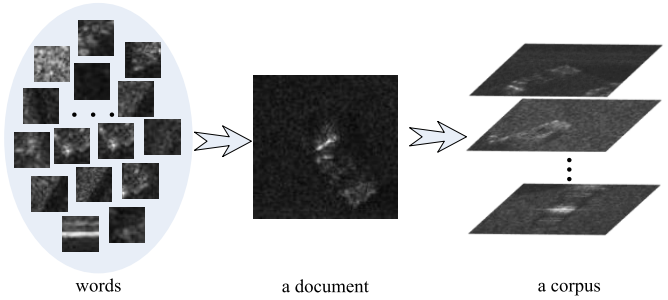


Fig. 3. Analogy between chip and text.

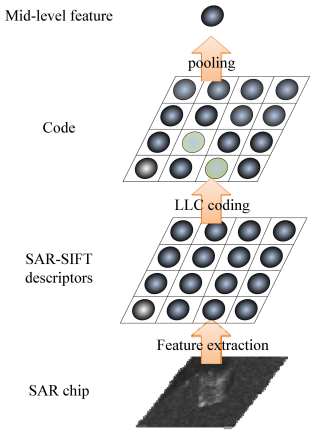


Fig. 4. Generation process of mid-level feature based on the BoVW model.

frequency of a document is turned into the K dimensional topic distribution to obtain the semantic information. In the text domain, words, which are directly extracted from the documents, can be considered as the low-level features of the documents. The word frequency, which is the statistical frequency of the low-level features, can be regarded as the mid-level feature of documents. The topic distribution learned by the LDA model can reflect the semantic information of documents and is regarded as the high-level feature.

B. Mapping Chip Into Text

Our purpose is to discriminate the target from the clutter in the SAR chips (small SAR images contain true targets or clutter false alarms) obtained by log-normal-based constant false alarm rate (CFAR) detector, which assumes that the background clutter obeys log-normal distribution, and target clustering method [46]. As we are trying to apply the technique used in the text domain into chips, there are two important issues that we have to address: 1) how to define correspondences of word, document, corpus, and codebook for chips and 2) how to construct the input, analogous to the word frequency in documents, of LDA model in chips. In the following, we present how these two issues are solved.

1) Analogy Between Chip and Text: In SAR target discrimination task, we define the following chip-text analogy and show the analogy in Fig. 3.

1) We partition a chip into patches by sliding a window over the chip with overlap. The patch is regarded as the basic processing unit and the low-level feature extracted from the patch is regarded as a word. The codebook is composed of the unique feature representation of patches.
2) A chip is a sequence of patches and regarded as a document.
3) A category is a collection of chips and equivalent to a corpus.

2) Mid-Level Feature Extraction From Chips Based on BoVW Model: In order to construct the input of LDA model in chips, we utilize the BoVW model [18], [12], [47] to represent each chip as the statistical frequency of low-level features, which is regarded as the mid-level feature of the chip. The BoVW model is a classical representation approach for image classification. It encodes the image with the statistical frequency of the contained words by the following three steps: low-level feature (word) extraction, low-level feature coding, and low-level feature pooling [18], [12], [47]. In our proposed SAR target discrimination approach, the generation process of mid-level feature based on the BoVW model is shown in Fig. 4, mainly containing the three steps: dense SAR-SIFT extraction, locality-constrained linear coding (LLC) model [48], and pooling, which correspond to the three steps in the BoVW model [18], [12], [47].

In our proposed approach, the SAR-SIFT descriptor is chosen as the low-level feature for its outstanding description ability of local spatial geometric characteristics and robustness to the speckle noise. We partition the chip into patches by sliding a window over the chip with overlap. In each patch,

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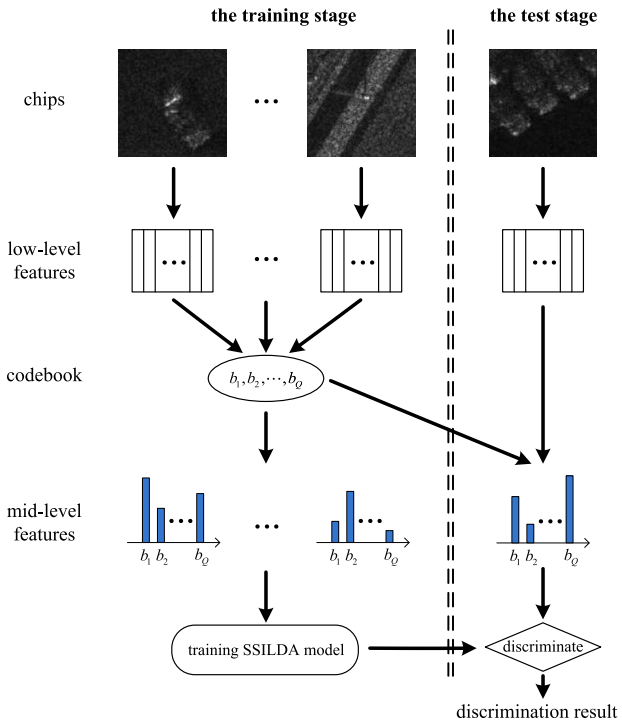


Fig. 5. Framework of our proposed semisupervised target discrimination method.

the SAR-SIFT descriptor is extracted. The extraction process of SAR-SIFT descriptor is the same as that in [17].

After the low-level feature extraction, all the extracted descriptors of the training samples are clustered to generate the codebook by using the k-means algorithm. The codebook is made up of Q cluster centers. Given the codebook, the LLC model is used to quantize each SAR-SIFT descriptor a Q-dimensional code with n nonzero elements. In our method, the LLC model is used since it has high computational efficiency and can obtain better reconstruction compared with the classical coding manners [48]. Moreover, we exploit the fast approximation of LLC [48] to implement the encoding.

After encoding the SAR-SIFT descriptors into Q-dimensional codes, all the codes inside each chip are pooled together by summing into a Q-dimensional vector to generate the mid-level feature of the chip. The mid-level feature is actually the statistical frequency of low-level local features.

By extracting the mid-level feature from the chip, we represent each chip as the statistical frequency of low-level features, analogous to the word frequency in the document. Thus, we can feed the mid-level feature into the modified LDA model to extract the high-level feature, which can reflect the semantic information of the chip.

III. SAR Target Discrimination Method Based on SSILDA

In this section, we present the semisupervised SAR target discrimination method based on the SSILDA model in detail. The flowchart of the proposed approach is shown in Fig. 5. After extracting the mid-level feature of the chip, we put it into our proposed SSILDA model to extract the high-level feature while implementing the discrimination with the extracted high-level feature. In the following, we introduce our proposed SSILDA model.

The original LDA model in [21] does not model clusters of documents. That is to say, all of the chips share the same Dirichlet prior α. In practice, different categories of chips have different prior over topics, while chips within the same class include a similar set of topics. Therefore, we introduce a class variable c into the model as the parent of the topic $\boldsymbol{\theta}$. In this way, each class defines a prior distribution in topic space, conditioned on which the topic probability vector $\boldsymbol{\theta}$ is sampled. Compared with the original LDA, we introduce a link between the class node c to each patch to model the intermediate topics for each class instead of sharing them among all classes. Moreover, due to the unsupervised nature of original LDA, it can only be developed for unsupervised image clustering. In order to implement classification, we incorporate some supervised information for class variable c to construct the semisupervised LDA, where a few chips are labeled and a large number of chips are unlabeled. By doing this, we combine the high-level semantic feature learning and the classifier learning into a uniform Bayesian framework. The joint learning of feature and classifier can avoid the unfitness of feature and classifier, thus guaranteeing the good discrimination performance.

In addition, in semisupervised LDA model, the number of topics is a significant parameter and has great influence on the discrimination performance. We introduce the DP into the semisupervised LDA model to automatically determine the number of topics. We call the modified semisupervised LDA model as SSILDA model for short. The graphical model of SSILDA is presented in Fig. 6, where the intuitional explanations of the model are shown at the end of the red and dashed arrows. The shaded node represents observed random variable, and the hashed fill node denotes partially observed random variable.

In our SSILDA model, the generative process of a chip can be described by Algorithm 2.

Algorithm 2 Generative Process of Chips in SSILDA

for each topic $k \in [1, \ldots, K]$ **do**
  Sample word mixture components $\varphi_k \sim Dir(\beta)$
end for
for each chip $j \in [1, \ldots, M]$ **do**
  Sample a class label $c_j \sim Mult(\boldsymbol{\eta})$
  Sample mixture proportion $\boldsymbol{\theta}_j \sim Dir(\boldsymbol{\alpha}_{c_j})$, $\boldsymbol{\alpha}_{c_j} \sim Dir(\gamma / K)$, $K \to \infty$
  for each patch $i \in [1, \ldots, N_j]$ in the chip j **do**
    Sample topic index $z_{ji} \sim Mult(\boldsymbol{\theta}_j)$
    Sample term for patch $w_{ji} \sim Mult(\varphi_{z_{ji}})$
  end for
end for

In order to understand the generative process for creating a chip in a specific class more directly, we put the process in plain English. First, we choose a class label, for example, the

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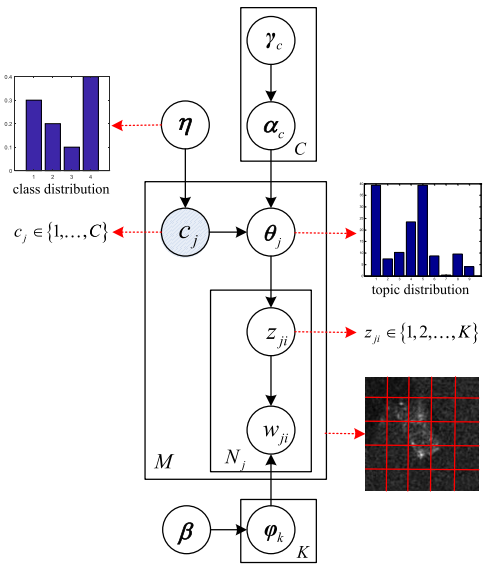


Fig. 6. Graphical model of SSILDA.

clutter (including buildings, trees, and roads). Given the clutter class, we draw a probability vector that will determine what intermediate topics to select when generating each patch of the chip. Now for creating each patch in the chip, we first determine a particular topic from the mixture of possible topics. For example, if a "building" topic is selected, this will in turn privilege some codewords that occur more frequently in buildings (e.g., dihedral angles). Now the topic favoring more dihedral angles is chosen, one can draw a codeword, which is likely to be a dihedral angle segment. The process of drawing both the topic and codeword is conducted many times, eventually forming an entire bag of patches that would construct a chip of buildings. In the following, we show the mathematical details of the formulation of the SSILDA model and how we learn its parameters.

As shown in Fig. 6, each chip is modeled as a mixture of K topics. In the original LDA model, K is assumed known, while in our SSILDA model, K is automatically determined by introducing the DP. Specially, we assume that the parameter α_c of mixture proportion is drawn from a Dirichlet distribution with parameter γ_c, i.e., $\alpha_c \sim \mathrm{Dir}(\gamma_c/K),\ K \to \infty$. From the topology of SSILDA, we can write the complete-data likelihood of a chip, i.e., the joint distribution of all known and hidden variables given the hyperparameters $\boldsymbol{\gamma}_c$, $\boldsymbol{\eta}$, and $\boldsymbol{\beta}$

$$\begin{aligned} &p(\boldsymbol{w}_j, \boldsymbol{z}_j, \boldsymbol{\theta}_j, \boldsymbol{c}_j, \boldsymbol{\alpha}_c | \boldsymbol{\gamma}_c, \boldsymbol{\eta}, \boldsymbol{\beta}) \\ &\quad = p(c_j|\boldsymbol{\eta}) p(\boldsymbol{\alpha}_{c_j}|\boldsymbol{\gamma}_{c_j}) p(\theta_j|\boldsymbol{\alpha}_{c_j}) \\ &\qquad \times \prod_{i=1}^{N} p(z_{ji}|\theta_j) p(w_{ji}|z_{ji}, \boldsymbol{\varphi}) p(\boldsymbol{\varphi}|\boldsymbol{\beta}) \end{aligned} \tag{1}$$

$$p(c_j|\boldsymbol{\eta}) = \mathrm{Mult}(c_j|\boldsymbol{\eta}) \tag{2}$$

$$p(\boldsymbol{\alpha}_{c_j}|\boldsymbol{\gamma}_{c_j}) = \mathrm{Dir}(\boldsymbol{\alpha}_{c_j}|\frac{\boldsymbol{\gamma}_{c_j}}{K}) \tag{3}$$

$$p(\theta_j|\boldsymbol{\alpha}_{c_j}) = \prod_{i=1}^{C} \mathrm{Dir}(\theta_j|\alpha_i)^{\delta(c_j,i)} \tag{4}$$

$$p(z_{ji}|\theta_j) = \mathrm{Mult}(z_{ji}|\theta_j) \tag{5}$$

$$p(w_{ji}|z_{ji}) = \prod_{k=1}^{K} p(w_{ji}|\varphi_k)^{\delta(z_{ji},k)} \tag{6}$$

$$p(\boldsymbol{\varphi}|\boldsymbol{\beta}) = \mathrm{Dir}(\boldsymbol{\varphi}|\boldsymbol{\beta}). \tag{7}$$

As shown in Fig. 6, the topic model is a hierarchical representation of the chip class model. The Dirichlet parameters $\boldsymbol{\alpha}_c$ and $\boldsymbol{\gamma}_c$ for each class are the class-level parameters, sampled once in the process of generating a class of chips. The multinomial variables θ_j are chip-level variables, sampled once per chip. Finally, the discrete topic variable z_{ji} and patch w_{ji} are patch-level variable, sampled every time a patch is generated. The marginal log likelihood of chip j is

$$\log p(\boldsymbol{w}_j|\boldsymbol{\gamma}_c, \boldsymbol{\beta}, \boldsymbol{\eta}) = \log \sum_{c_j=1}^{C} p(c_j|\boldsymbol{\eta}) p(\boldsymbol{\alpha}_{c_j}|\boldsymbol{\gamma}_{c_j}) p(\boldsymbol{w}_j|\boldsymbol{\alpha}_{c_j}, \boldsymbol{\beta}). \tag{8}$$

Compared with the original LDA, we introduce a link between the class node c to each chip to model the intermediate topics for each class instead of sharing them among all classes, such that $w_{ji} \sim p(w_{ji}|z_{ji}, \boldsymbol{\beta}, c)$. The generative equations above are hence changed according to this dependency on c. Moreover, we employ the DP on the topic mixture proportion parameter $\boldsymbol{\alpha}_c$ to automatically determine the number of topic. This allows the number of topic to be determined based on the data, rather than preset its value.

A. Learning: Collapsed Gibbs Sampling

In the SSILDA model, the topic mixture proportion $\boldsymbol{\theta}_j \sim \mathrm{Dir}(\boldsymbol{\alpha}_{c_j})$ and its parameter $\boldsymbol{\alpha}_{c_j} \sim \mathrm{Dir}(\gamma/K),\ K \to \infty$. Since the Dirichlet is not conjugate to itself, the parameters cannot be inferred by variable Bayesian (VB). Therefore, we derive a collapsed Gibbs sampler to infer the model's parameters. The hidden variables of our model are c_j, the class which the chip belongs to, and z_{ji}, the topic that appears with the word w_{ji}. We do not need to include, i.e., can integrate out, the parameter sets $\boldsymbol{\theta}$ and $\boldsymbol{\varphi}$ because they can be interpreted as statistics of the associations between the observed w_{ji} and the corresponding z_{ji}. Moreover, we keep the parameter $\boldsymbol{\alpha}_c$ in the DP directly and derive its posterior. In our model, given parameters $\boldsymbol{\alpha}_c$, $\boldsymbol{\eta}$, and $\boldsymbol{\beta}$ and the joint distribution of $\boldsymbol{w}$, $\boldsymbol{z}$, and $\boldsymbol{c}$ can be factored:

$$p(\boldsymbol{w}, \boldsymbol{z}, \boldsymbol{c}|\boldsymbol{\alpha}_c, \boldsymbol{\beta}, \boldsymbol{\eta}) = p(\boldsymbol{w}|\boldsymbol{z}, \boldsymbol{\beta}) p(\boldsymbol{z}|\boldsymbol{\alpha}_c, \boldsymbol{c}) p(\boldsymbol{c}|\boldsymbol{\eta}). \tag{9}$$

Analogous to the deduction in [49], from the joint distribution, we can derive the conditional posterior of the topic z_{ji}

$$\begin{aligned} &p(z_{ji} = k|z_{-ji}, \boldsymbol{w}, \boldsymbol{c}) \\ &\quad \propto \frac{n_{k,-ji}^{(t)} + \beta_t}{\sum_{t=1}^{Q} (n_{k,-ji}^{(t)} + \beta_t)} \cdot \frac{n_{j,-ji}^{(k)} + \alpha_{ck}}{\sum_{k=1}^{K} (n_{j,-ji}^{(k)} + \alpha_{ck}) - 1} \end{aligned} \tag{10}$$

where $n_{k,-ji}^{(t)}$ denotes the number of times the tth word occurs with kth topic with the exception of the value on site i in chip j. Likewise, $n_{j,-ji}^{(k)}$ denotes the number of times kth topic occurs in jth chip with the exception of the value on site i in chip j. Note that the sampling space of z_{ji} has $K+1$ dimensions, with the last point mass $\alpha_{c,K+1}/Q$,

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which represents the unseen components. Whenever this mass is sampled, a new topic is created.

In addition, the labeled chips are allocated user-provided class labels. For the jth chips, the conditional posterior of the class c_j can be given by

$$p(c_j|\boldsymbol{w},\boldsymbol{z}) \propto \eta_c \cdot \frac{\Delta(\boldsymbol{n}_j + \boldsymbol{\alpha}_{c_j})}{\Delta(\boldsymbol{\alpha}_{c_j})}, \quad \boldsymbol{n}_j = \left\{n_j^{(k)}\right\}_{k=1}^{K}. \tag{11}$$

It is worth noting here that the class label of a chip should be kept untouched if the chip corresponds to the labeled training chip. Otherwise, the chip will be allocated a new class label by sampling using (11).

The sampling process of the topic z_{ji} and class c_j is repeated until the maximum number of iterations has been reached. The final sampling results are used to approximate the parameters of chip topic mixture proportion and word mixture proportion:

$$\theta_{j,k} = \frac{n_j^{(k)} + \alpha_{c_j k}}{\sum_{k=1}^{K}\left(n_j^{(k)} + \alpha_{c_j k}\right)} \tag{12}$$

$$\varphi_{k,t} = \frac{n_k^{(t)} + \beta_t}{\sum_{t=1}^{Q}\left(n_k^{(t)} + \beta_t\right)}. \tag{13}$$

The prior proportions α_c can be sampled by simulating how new components are created for n_{jk} draws from the DP with precision α_{ck}, which is a sequence of Bernoulli trials for each j and k [50]

$$p(m_{jkr} = 1) = \frac{\alpha_{c_j k}}{\alpha_{c_j k} + r - 1} \quad \forall r \in \left[1, n_j^{(k)}\right], \quad j \in [1, M], \ k \in [1, K]. \tag{14}$$

A posterior sample of the root distribution then is obtained via

$$\boldsymbol{\alpha}_c \sim \mathrm{Dir}\{\{m_k\}_k, \gamma_c\} \text{ with } m_k = \sum_j \sum_r m_{jkr}. \tag{15}$$

We assume the precision parameter γ_c of the DP as gamma prior by introducing hyper-prior parameters a_{γ_c} and b_{γ_c}. Auxiliary variables u_c and v_c are introduced for the DP gamma prior that are governed by Bernoulli and Beta distributions. The parameter γ_c is then sampled using the gamma distribution

$$u_c \sim \mathrm{Bern}\left(\frac{T}{T + \gamma_c}\right), v_c \sim \mathrm{Beta}(\gamma_c + 1, T) \tag{16}$$

$$\gamma_c \sim \mathrm{Gam}(a_{\gamma_c} + K - 1 + u_c, b_{\gamma_c} - \log v_c) \tag{17}$$

where $T = \sum_k m_k$ is the augmented variable of $\boldsymbol{\alpha}_c$.

B. Bayesian Decision

After obtaining the model parameters, the next step is to make decisions given an unknown chip. An unknown chip is first represented by the mid-level feature. Given the chip representation $\boldsymbol{w}_j$, we can inference the probability of each class using (11). Then, the decision of class is made by comparing the probability of each class: $c_j = \arg\max_{c_j} q(c_j|\gamma_{jc_j}, \phi_{jc_j})$.

IV. Experimental Results and Analysis

This section presents the experimental results on two measured data sets to illustrate the validity and popularity of our proposed SSILDA model for SAR target discrimination. First, the two SAR image real data sets used in our experiments and the experimental setup are introduced. Then, the proposed SSILDA model is analyzed in detail and compared with the original LDA model. Finally, quantitative results of our proposed method, supervised, and other existing classical semisupervised methods are exhibited and analyzed, which demonstrates the advantages of our proposed method.

A. Experimental Setup

To validate the effectiveness of the target discrimination based on our SSILDA model, experiments are performed on the miniSAR data set and FARAD SAR data set[1] acquired by Sandia National Laboratories of America, Albuquerque, NM, USA, in 2005 and 2015, respectively. The miniSAR data set, with 4-in resolution and size 1638×2501, totally contains 20 SAR images, of which some are without targets of interest (like cars and aircrafts) and some are with simple scenes. We select seven typical SAR images with complex scenes and targets of interest from the miniSAR data set for our experiments, where three images are used as the training data and the remaining four images are used as the test data. A SAR image from the miniSAR data set is shown in Fig. 7(a), and the cars are marked with red rectangles. We also show the corresponding optical image from Google Earth in Fig. 7(b). Since the SAR image and optical image are acquired at different times and angles, they are not completely corresponding. The FARAD SAR data set, with 4-in resolution, totally contains 175 SAR images. In these images, some are without targets and repeated. We select seven images with targets and complex scenes from FARAD SAR data set to carry out our experiments, in which four images are used as the training data and the remaining three images are used as the test data. Two SAR images used in our experiments from the FARAD SAR data set are shown in Fig. 8, and the cars are marked with red rectangles.

From Figs. 7 and 8, we can see that the scenes of SAR images in miniSAR data set and FARAD data set are rather complex, including cars, buildings, roads, grasslands, trees, and streetlights, in which the cars are the targets of interest. The complex scenes of SAR images bring great challenge to the target discrimination. Especially, this challenge is extraordinarily obvious for the FARAD SAR data set due to the close-packed buildings, shelter from trees, and great similarity between targets and clutter.

The log-normal-based CFAR, which assumes that the background clutter obeys log-normal distribution, and target clustering method [46] are first utilized to obtain chips from SAR images of miniSAR data set and FARAD SAR data set. The size of chip is 100×100. The detection results of the seven SAR images in miniSAR data set are given in Table I, where the detected chips of three images are regarded as the training

[1] https://www.sandia.gov/radar/complex-data/index.html

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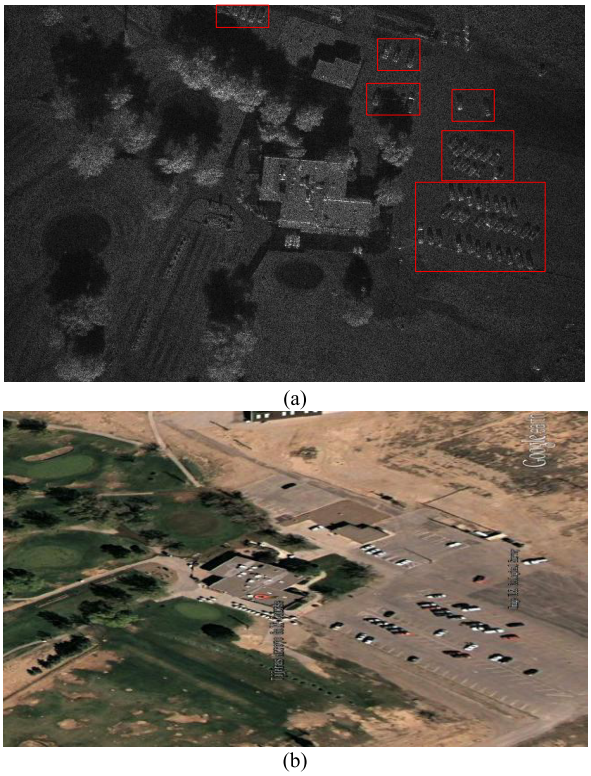


Fig. 7. Example of the data. (a) SAR image from the miniSAR data. (b) Corresponding optical image from Google Earth.

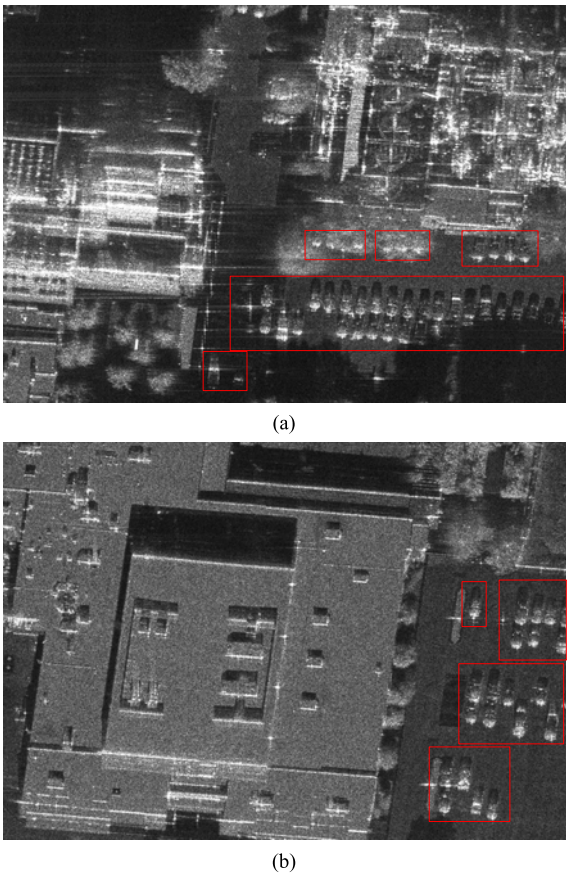


Fig. 8. Two SAR images of the FARAD SAR data set. (a) Training image. (b) Test image.

TABLE I
NUMBER OF CHIPS DETECTED FROM SEVEN SAR IMAGES IN THE MINISAR DATA SET

	Number of total chips	*Number of targets*	*Number of clutters*
Training set	*469*	*156*	*313*
Test set	*426*	*153*	*273*

TABLE II
NUMBER OF CHIPS DETECTED FROM SEVEN SAR IMAGES IN THE FARAD SAR DATA SET

	Number of total chips	*Number of targets*	*Number of clutters*
Training set	*399*	*157*	*242*
Test set	*220*	*66*	*154*

set and the detected chips of the remaining four images are regarded as the test set. Table II lists the detection results of the seven SAR images in the FARAD SAR data set, where the detected chips of four images are regarded as the training set and the detected chips of the remaining three images are regarded as the test set. Table III gives some chips of miniSAR data set and FARAD SAR data set, including targets and clutter. In the following experiments, in order to reduce the influence of imbalanced positive and negative training data to the discrimination performance, in the training set, we randomly select negative samples with the same size as the positive samples from all the clutter chips.

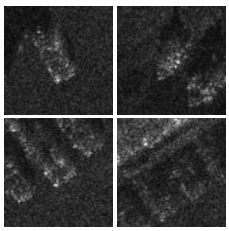
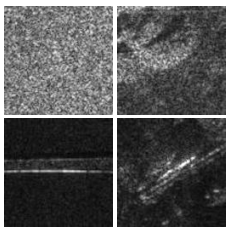
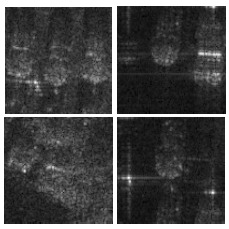
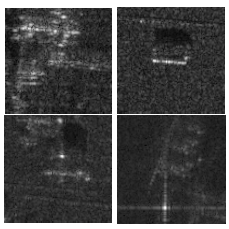
B. Analysis of the SSILDA Model

In our proposed method, the SSILDA model combines the feature learning and classifier learning into a uniform Bayesian framework. In this section, we analyze the SSILDA model in detail. Target discrimination is essentially a binary classification problem, whose main task is to discriminate the targets from clutter. Therefore, the class number C in SSILDA is equal to 2.

1) Analysis of Patch Size and Codebook Size: When partitioning the chip into patches in the mid-level feature extraction stage, the size of the patch will affect the final discrimination performance. If the patch size is too small, the local information of extracted SAR-SIFT descriptor is insufficient. If the patch size is too large, the SAR-SIFT descriptor cannot reflect the local spatial information of a patch. We change the size of a patch as 8×8, 16×16, 32×32, and 64×64 to carry out experiments on the two SAR image data sets. In these experiments, when learning the SSILDA model, the number of labeled samples is set as 18, where the number of labeled

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TABLE III
SOME CHIP EXAMPLES AFTER THE DETECTION STAGE

Dataset	Target chips	Clutter chips
miniSAR		
FARAD SAR		

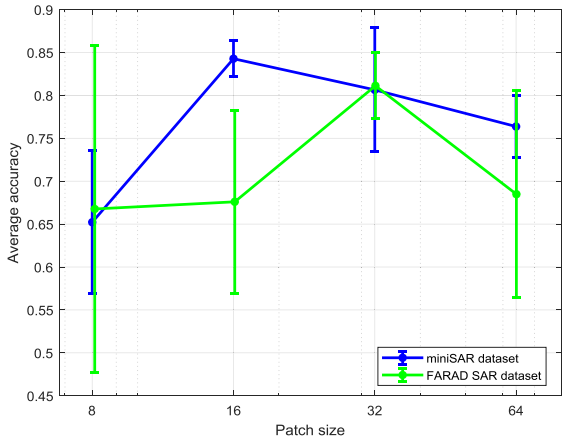


Fig. 9. Variation of average accuracy and standard deviation with the patch size.

positive samples is equal to that of labeled negative samples. The results are averaged over ten repetitions and for each repetition, labeled samples are randomly selected. Moreover, the space between two patches is set as 4 in these four cases. Fig. 9 shows the variation curve of average accuracy and the corresponding standard deviation for the test SAR images with different sizes of a patch. As shown in Fig. 9, for the miniSAR data set, when the size of patch is 16×16, the average accuracy reaches its maximum. For the FARAD SAR data set, when the size of patch is 32×32, the average accuracy reaches its maximum. Therefore, in the following experiments, we set the patch size as 16×16 for the miniSAR data set and 32×32 for the FARAD SAR data set.

The codebook size Q is another important parameter for the SSILDA model. The same with the patch size, we present the variation of average accuracy and standard deviation on miniSAR and FARAD SAR data sets with different values of

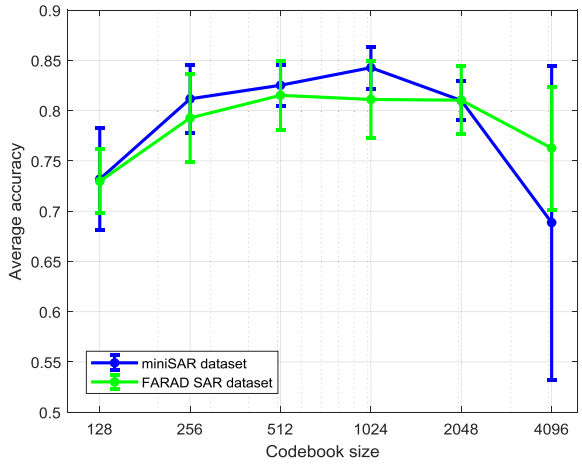


Fig. 10. Variation of average accuracy and standard deviation with the codebook size.

the codebook size Q in Fig. 10. In this experiment, the number of labeled samples is also set as 18, where the number of labeled positive samples is equal to that of labeled negative samples. The results are averaged over ten repetitions and for each repetition, labeled samples are randomly selected. From Fig. 10, we can see that the discrimination performance first increases and then descends as the size of codebook increases.

For the miniSAR data set, when the size of codebook is equal to 1024, the discrimination performance is the best. For the FARAD SAR data set, when the size of codebook is equal to 512, 1024, and 2048, the discrimination performance is comparable. However, the smaller the codebook size is, the lower the computational cost is. Therefore, we set the codebook size as 1024 for the miniSAR data set and 512 for the FARAD SAR data set in all the following experiments.

2) Comparison With the Original LDA Model: The main difference between our SSILDA model and the original LDA model is that we introduce the class node in the SSILDA. The advantages of introducing the class node are twofold. On the one hand, the introduction of class node makes that the chips from different classes have different topic priors, thus enhancing the separability between the posteriors of topic distributions from different classes. On the other hand, the original LDA model is unsupervised and can be only used for clustering and feature extraction. By introducing the class node and adding supervised information of partial samples, we can implement the joint learning of feature and classifier. The unfitness of features and discriminator can be avoided, thus guaranteeing the good discrimination performance. In Fig. 11, we give a target chip and its posterior topic distributions learned by the original LDA model and SSILDA model. In Fig. 12, we give a clutter chip and its posterior topic distributions learned by the original LDA model and SSILDA model. In Table IV, we give the ratio of between-class distance to within-class distance (BWR) of all the test samples' posterior topic distributions learned by the original LDA model and SSILDA model to quantitatively compare their linear separability. The BWR is defined as follows:

$$\mathrm{BWR} = L_{\mathrm{between}}/L_{\mathrm{within}} \tag{18}$$

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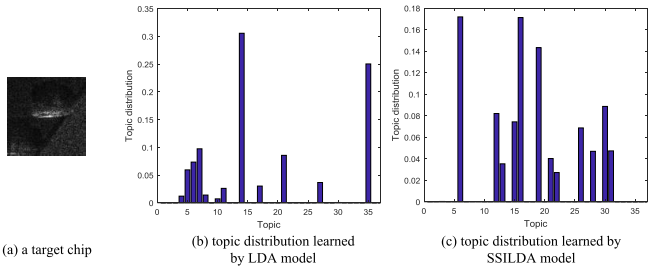


Fig. 11. (a) Target chip and its topic distributions learned by the (b) LDA model and (c) SSILDA model.

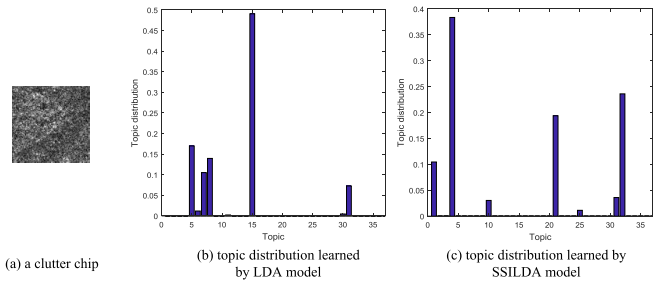


Fig. 12. (a) Clutter chip and its topic distributions learned by the (b) LDA model and (c) SSILDA model.

TABLE IV
BWR VALUES OF ALL TEST SAMPLES' POSTERIOR TOPIC DISTRIBUTIONS LEARNED BY THE ORIGINAL LDA MODEL AND SSILDA MODEL

	Topic distribution of LDA model	Topic distribution of SSILDA model
BWR	1.1016	1920

$$L_{\text{within}} = \sum_{p=1}^{P} \frac{1}{n_p} \sum_{k}^{n_p} \left(x_k^{(p)} - \bar{x}^{(p)}\right)^T \left(x_k^{(p)} - \bar{x}^{(p)}\right) \quad (19)$$

$$L_{\text{between}} = \sum_{p=1}^{P} \left(\bar{x}^{(p)} - \bar{x}\right)^T \left(\bar{x}^{(p)} - \bar{x}\right) \quad (20)$$

where BWR is the BWR value of feature $\{x_k^{(p)}\}_{p=1,k=1}^{P,n_q}$, L_{between} represents the between-class distance in the feature space, and L_{within} represents the within-class distance in the feature space. P denotes the number of classes, which is 2 in our experiments. n_p denotes the number of samples from the pth class and $x_k^{(p)}$ denotes the kth sample from the pth class. $\bar{x}^{(p)}$ is the mean value of all samples from the pth class, and $\bar{x}$ represents the mean value of samples from all the classes. The smaller L_{within} and the larger L_{between} are in the feature space, the larger the BWR value of the feature is, which means the feature is more linearly separable. From Table IV, we can see that the BWR value of posterior topic distribution learned by SSILDA is higher than that learned by the original LDA, which demonstrates that the posterior topic distribution learned by the SSILDA has better linear separability. This verifies the effectiveness of introducing the class node into the LDA model.

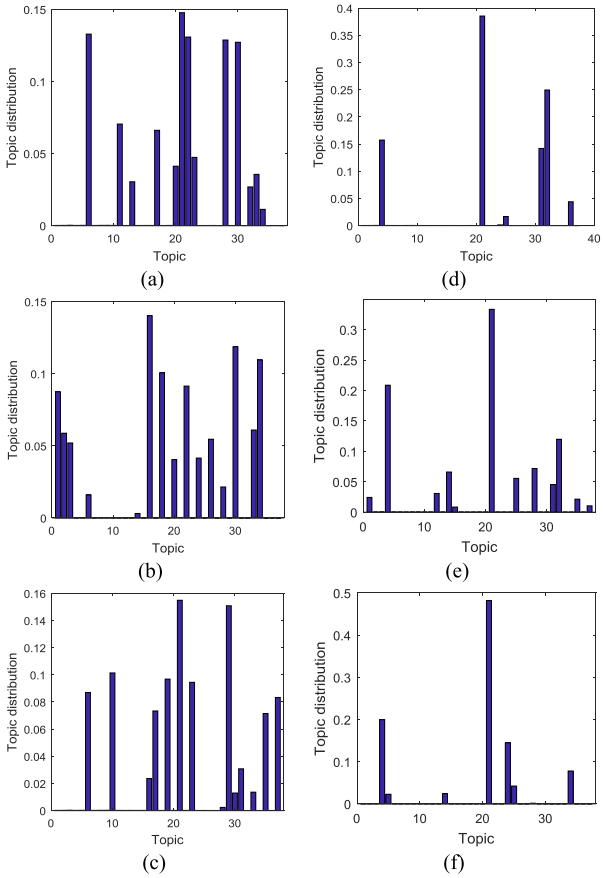


Fig. 13. Topic distribution examples of target and clutter chips. (a)–(c) Topic distributions of three target chips. (d)–(f) Topic distributions of three clutter chips.

3) Analysis of High-Level Feature: In our SSILDA model, the posterior topic distribution of a chip can be used as the feature for target discrimination. As has been mentioned in Section II-B, the posterior topic distribution of a chip can reflect the semantic information of the chip and can be regarded as the high-level feature. Fig. 13 shows the posterior topic distributions of some target and clutter chips. Fig. 13(a)–(c) shows the posterior topic distributions of target chips, and Fig. 13(d)–(f) shows the posterior topic distributions of clutter chips. From Fig. 13, we can see that the target chips have large values in most topics, while the clutter chips have large values only in few topics, which illustrates that the topic distributions of targets and clutter have large difference. This demonstrates the effectiveness of using the posterior topic distribution as feature to discriminate targets from clutter.

Moreover, we quantitatively compare the linear separability of classical Lincoln features, mid-level feature, and the high-level feature by BWR. The BWR values of classical Lincoln features [8], mid-level feature, and our high-level feature on the miniSAR data set, all extracted from the test samples, are listed in Table V. The results in Table V indicate that the BWR value of our high-level feature is larger than that of classical Lincoln features and mid-level feature. In addition,

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TABLE V
BWR VALUES OF LLC FEATURE AND HIGH-LEVEL FEATURE

	High-level feature	Mid-level feature	Lincoln features
BWR	1.1920	1.0355	1.0321

TABLE VI
AVERAGE ACCURACY AND STANDARD DEVIATION OF LINEAR SVM WITH DIFFERENT FEATURES

	High-level feature + linear SVM	Mid-level feature + linear SVM	Lincoln features + linear SVM
Average accuracy (std)	0.8492(0.0142)	0.8454 (0.0209)	0.8319 (0.008)

we also compare the discrimination performance of linear SVMs with high-level feature, mid-level feature, and Lincoln feature on the miniSAR real SAR data. Table VI gives the discrimination results of these three methods. From Table VI, it can also be seen that the discrimination performance of our high-level feature is better than those of mid-level feature and Lincoln features. Results of Tables V and VI demonstrate that the high-level feature extracted via our SSILDA model can linearly discriminate targets from clutter more easily than the classical Lincoln features and mid-level feature.

4) Analysis of Joint Learning of Feature and Discriminator: As described in Section II, most of the existing SAR target discriminators are independently designed from the features. However, it may occur that the features are not suitable for the classifier, and in this case, the discrimination performance will degrade. In order to avoid this problem, we propose to combine the feature learning and discriminator learning into a uniform Bayesian framework. To demonstrate the advantage of combining the feature learning and discriminator learning, we compare our proposed method with "high-level feature + linear SVM/radial basis kernel SVM/polynomial kernel SVM" on the miniSAR data set. In "high-level feature + linear SVM/radial basis kernel SVM/polynomial kernel SVM," the high-level feature is learned with our SSILDA model and all the training samples are used to train the SVM. Moreover, for the fair comparison, all the training samples are labeled and used for learning the SSILDA model. Table VII shows the average accuracy and corresponding standard deviation of different methods. The results in Table VII are averaged over ten choices of different labeled samples. For the sake of brevity, the radial basis kernel SVM is denoted by RSVM, and the polynomial kernel SVM is denoted by PSVM in Table VII. From Table VII, we can see that when the high-level feature is put into different types of classifiers, the discrimination performance will have great differences. When there exists mismatch between the feature and discriminator, for example, high-level feature with RSVM and PSVM, the discrimination performance will degrade. In our proposed method, by jointly learning the feature and classifier, the good discrimination performance can be assured.

TABLE VII
AVERAGE ACCURACY AND STANDARD DEVIATION OF DIFFERENT METHODS

	Our proposed method	High-level feature + linear SVM	High-level feature + RSVM	High-level feature + PSVM
Average accuracy (std)	0.8567 (0.0099)	0.8508 (0.0107)	0.5093 (0.0152)	0.6358 (0.1594)

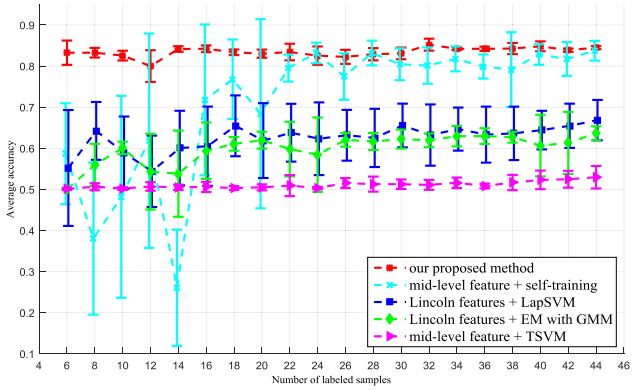
C. Comparison With Existing Methods

1) Comparison With Classical Semisupervised Methods: In order to illustrate the performance of our proposed semisupervised discrimination method, we first compare it with four classical semisupervised methods, including self-training, TSVM [39], EM with GMM [40], and LapSVM [41]. For self-training algorithm, the linear SVM is used as the base classifier. A set of experiments with different numbers of labeled samples are conducted, where the number of labeled positive samples is equal to that of labeled negative samples. Fig. 14 shows the variations of average accuracy and the corresponding standard deviation via different semisupervised methods with the number of labeled samples. Fig. 14(a) gives the results on the miniSAR data set, and Fig. 14(b) gives the results on the FARAD SAR data set. The results in Fig. 14 are averaged over ten repetitions and for each repetition, the labeled samples are randomly selected. It should be noted that we use both the Lincoln features and mid-level feature as the input of the classical semisupervised methods, and Fig. 14 only shows the best results.

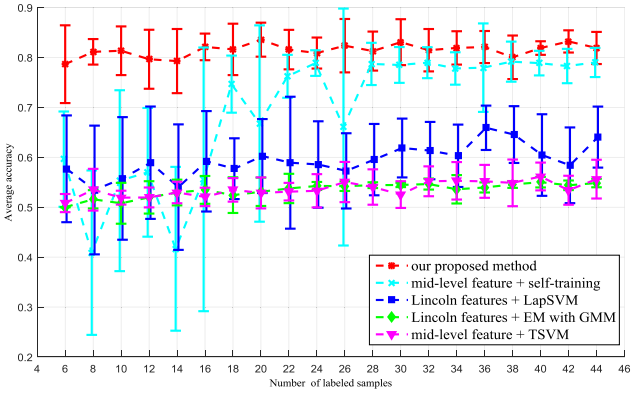
Generally speaking, these five semisupervised methods work better and better when the number of labeled samples increases. However, it does not mean that the more the labeled samples we use, the better the performance is. It is because that the discrimination performance depends not only on the number of labeled samples but also on the useful information provided by the labeled samples. Moreover, it can be seen that our proposed method outperforms the existing classical semisupervised methods. There are mainly two reasons: 1) our proposed SSILDA model can mine the underlying semantic information in samples and 2) our proposed method combines the feature learning and classifier learning into a uniform framework, avoiding the unfitness between feature and classifier while making the labeled information contribute not only the classifier learning but also the feature learning. Furthermore, it is noted that even with few labeled samples, our proposed method can obtain good discrimination performance. This demonstrates the effectiveness of our proposed semisupervised method in SAR target discrimination.

2) Comparison With Supervised Discrimination Methods: To further demonstrate the advantages of our proposed discrimination method, we compare it with the existing classical

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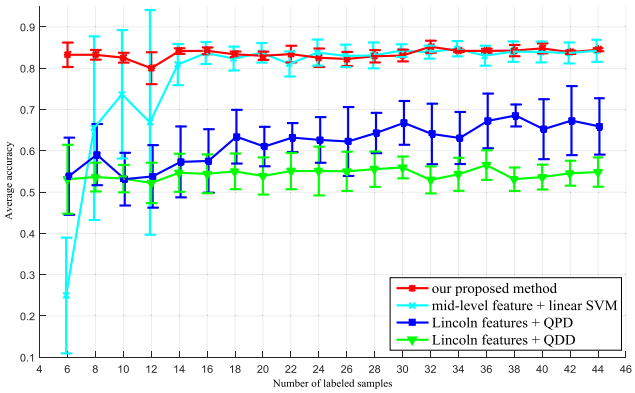


(a) miniSAR

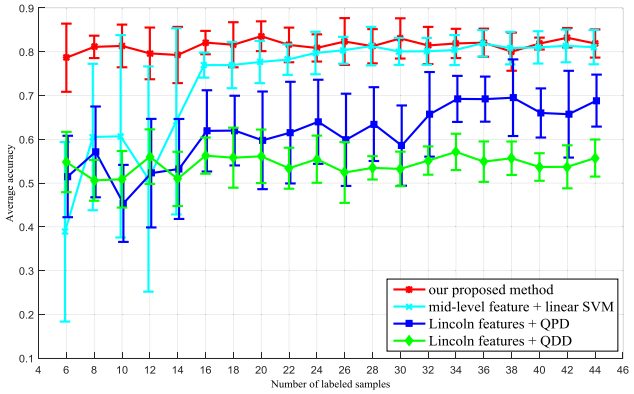


(b) FARAD SAR

Fig. 14. Variations of average accuracy and corresponding standard deviation via different semisupervised methods with the number of labeled samples. (a) Results on the miniSAR data set. (b) Results on the FARAD SAR data set.



(a) miniSAR



(b) FARAD SAR

Fig. 15. Variations of average accuracy and corresponding standard deviation via different supervised methods with the number of labeled samples. (a) Results on the miniSAR data set. (b) Results on the FARAD SAR data set.

supervised discrimination methods, including linear SVM, QPD [30], and QDD [28]. A set of experiments with different numbers of labeled samples are conducted, where the number of labeled positive samples is equal to that of labeled negative samples. Fig. 15 shows the variations of average accuracy and the corresponding standard deviation via different supervised methods with the number of labeled samples, where Fig. 15(a) shows the results on the miniSAR data set, and Fig. 15(b) shows the results on the FARAD SAR data set. The results in Fig. 15 are averaged over ten repetitions, and for each repetition, the labeled samples are randomly selected. Moreover, we also used the Lincoln features and mid-level feature as the input of the supervised classifiers, and Fig. 15 only gives the best results. From Fig. 15, we can see that, in general, our proposed discrimination method is superior to the classical supervised discrimination methods and has stable performance. Especially, the discrimination performance of our proposed method is far better than those of "Lincoln features + QPD" and "Lincoln features + QDD." For "mid-level feature + SVM," when there are only a few labeled samples, its discrimination performance is considerably bad. This demonstrates that by leveraging the labeled samples and unlabeled samples, the discrimination performance can be improved. Thus, the difficulty in SAR target discrimination that large number of labeled samples is needed to obtain good performance can be solved.

V. Conclusion

In this article, we propose a new target discrimination method based on the SSILDA model. First, the mid-level feature is extracted from the chips to obtain the structure information. Then, we extend the LDA model by incorporating the information of a few of labeled samples and large number of unlabeled samples to combine the feature learning and classifier learning into a uniform Bayesian framework. Moreover, the DP is introduced into the extended LDA model to automatically determine the number of topics. Note that the lack of supervised information is common in operational applications of SAR target discrimination. In such situation, the combination of labeled and unlabeled samples in a semisupervised framework can significantly improve the discrimination performance. Experimental results on the measured data sets show that our proposed method outperforms the existing classical semisupervised methods with the same size of labeled samples. Moreover, our method can obtain better performance than the traditional supervised discrimination methods based on Lincoln features and mid-level feature, which demonstrates the effectiveness of our proposed method.

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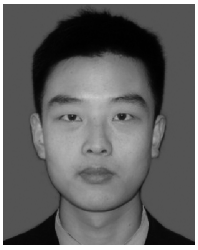
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